

S.T.A.R. Research Charter

Scientific Specification for Arithmetic–Geometric–Topological Investigation of Cosmic Structure

Project: S.T.A.R. Program

Status: Research Charter v0.2 — Foundational Specification

Purpose: Canonical scientific specification for the S.T.A.R. research program

Primary domains: Arithmetic geometry, topology, topological data analysis, astrophysics, cosmology, statistical inference, mathematical physics, machine learning

Controlling Scientific Standard for the Arithmetic–Geometric–Topological Investigation of Cosmic Structure

1. Purpose

The S.T.A.R. program investigates whether mathematical structures arising from arithmetic geometry can be mapped into geometric and topological structures in a manner that produces statistically nontrivial, reproducible, and independently testable correspondence with astronomical observations.

The central research question is:

Can arithmetic invariants be mapped into geometric and topological structures in a way that produces statistically nontrivial, reproducible correspondence with independent astronomical data?

The program is therefore an investigation of a possible correspondence between:

[
 \text{Arithmetic}
 \rightarrow
 \text{Geometry}
 \rightarrow
 \text{Topology}
 \rightarrow
 \text{Observation}]

\rightarrow
\text{Dynamics}
\rightarrow
\text{Physics}.
]

The project does **not** assume that elliptic curves determine the universe.

In particular, S.T.A.R. does not adopt any of the following as established facts:

- that elliptic curves determine cosmic structure;
- that elliptic-curve rank is equivalent to physical time;
- that elliptic-curve rank is equivalent to gravity;
- that arithmetic structure replaces the Einstein field equations;
- that dark matter is arithmetic curvature;
- that dark energy is rank growth;
- that cosmic topology is identical to arithmetic topology;
- that a unique arithmetic-to-cosmic mapping exists;
- that a strict bijection between arithmetic objects and astronomical objects exists;
- that the Birch–Swinnerton-Dyer conjecture is required for the research program;
- that historical numerical correlations establish a physical law.

These remain hypotheses, candidate models, conjectures, or speculative interpretations unless and until they satisfy the evidentiary standards defined by this Charter.

The Charter is the controlling standard for evaluating all subsequent S.T.A.R. claims, experiments, mappings, computational results, theoretical constructions, and proposed physical interpretations.

2. Scientific Scope

S.T.A.R. consists of several connected but epistemically distinct research layers.

Layer I — Arithmetic

The program studies mathematical objects and invariants including elliptic curves, rank, conductor, discriminant, regulator, torsion, periods, Frobenius traces, and related quantities.

Layer II — Arithmetic Geometry

Arithmetic invariants are transformed into candidate geometric representations through mappings such as:

$(\phi_E, \theta_E, z_E, s_E).$
]

A simplified candidate representation is:

$\left(\begin{array}{l} \log|\Delta_E|, \\ \log N_E, \\ \alpha_{r_E} \end{array} \right).$
]

Layer III — Arithmetic Cartography

Mapped arithmetic objects are treated as points, fields, densities, graphs, surfaces, or other geometric constructions.

Layer IV — Topology

The resulting structures are studied using persistent homology, persistence diagrams, persistence landscapes, persistence images, Wasserstein distances, bottleneck distances, and related topological tools.

Layer V — Astronomical Comparison

Arithmetic-derived structures are compared with independent astronomical data, including:

- galaxy distributions;
- clusters;
- filaments;
- voids;
- matter density;
- galaxy morphology;
- gravitational observables;
- radiation;
- redshift;
- galaxy environment;
- star formation rate;
- gas mass;
- stellar mass;
- other independently measured astronomical quantities.

Layer VI — Dynamics

Where static correspondence is supported, the program investigates whether arithmetic structures can describe or parameterize observed evolution across redshift or other observational variables.

Layer VII — Physical Theory

Only after empirical correspondence has been independently established may the program investigate whether an arithmetic structure can be incorporated into a physical theory involving:

- fields;
- gravity;
- thermodynamics;
- entropy;
- cohomology;
- scalar fields;
- variational principles;
- conservation laws;
- effective field theory;
- cosmological dynamics.

This ordering is mandatory.

A speculative physical interpretation must not be used as evidence for the empirical correspondence that motivated it.

3. Epistemic Classification System

Every substantive S.T.A.R. statement must be assigned an epistemic category.

The following categories are canonical.

Code	Category	Meaning
D	Derived	Follows mathematically from stated assumptions or definitions
E	Empirical	Directly supported by measured or computationally reproducible evidence
M	Model	A mathematical or phenomenological construction introduced for investigation
P	Postulate	An explicitly adopted assumption of a proposed theory

H	Hypothesis	A proposition being experimentally tested
C	Conjecture	A broader mathematical or physical proposition lacking proof
T	Testable Prediction	A quantitative or qualitative consequence that can be tested independently
S	Speculative Interpretation	A possible interpretation not yet supported sufficiently to be treated as a hypothesis or empirical conclusion

These classifications are not interchangeable.

For example:

“Rank corresponds to cosmic complexity”

is a **H**.

The statement:

“The model predicts that rank distributions differ between clusters and voids”

is a **T**.

The statement:

“A measured dataset contains a statistically significant difference”

may be **E**, provided the analysis is reproducible and satisfies the statistical requirements of this Charter.

The statement:

“Therefore rank is the physical origin of cosmic structure”

remains **S** unless substantially stronger evidence and theory are established.

4. Core Mathematical Objects

4.1 Elliptic Curves

An elliptic curve over $(\mathbb{Q})$ is treated as a mathematical object:

[
 $E/\mathbb{Q}$.
]

Relevant invariants may include:

[
 r_E ,
 $\quad$
 N_E ,
 $\quad$
 Δ_E ,
 $\quad$
 $\operatorname{Reg}(E)$,
 $\quad$
 T_E ,
]

together with other invariants where mathematically appropriate.

No individual invariant is assumed a priori to be physically privileged.

4.2 Rank

The Mordell–Weil rank

[
 r_E
]

is one candidate arithmetic organizing variable.

S.T.A.R. therefore tests propositions such as:

[
 $P(r_E \mid \text{cluster})$,
 $\quad$
 $P(r_E \mid \text{filament})$,
 $\quad$
 $P(r_E \mid \text{void})$.
]

Rank is not defined by this Charter to be equivalent to any astronomical quantity.

4.3 Candidate Mapping

A candidate mapping may take the form

$$\begin{bmatrix} \Phi: \\ \mathcal{E} \\ \rightarrow \\ \mathcal{X}, \\ \end{bmatrix}$$

where ($\mathcal{E}$) is a set of arithmetic objects and ($\mathcal{X}$) is a geometric representation space.

A historical candidate mapping is:

$$(\phi_E, \theta_E, z_E, s_E),$$

with quantities derived from arithmetic invariants.

A simplified historical form is:

$$\begin{bmatrix} \left(\right. \\ \log|\Delta_E|, \\ \log N_E, \\ \alpha_{r_E} \\ \left. \right). \\ \end{bmatrix}$$

The mapping is a **projection**, not necessarily a bijection.

Different arithmetic objects may therefore map to equivalent or nearby geometric states.

No requirement of one-to-one correspondence shall be imposed unless an experiment specifically tests that property.

5. Candidate Mapping Families

S.T.A.R. retains multiple mapping families to prevent dependence on one historically selected construction.

These include:

5.1 (Φ)

The primary historical arithmetic-invariant mapping family.

5.2 MCJ

A mapping involving the (j)-invariant or related modular information.

5.3 PTD

A predominantly algebraic mapping that minimizes dependence on analytic constructions.

5.4 FT

A Frobenius-trace-based mapping using finite-field information.

These alternative families are scientifically important because a correspondence that exists only under one specially selected mapping may be consistent with model-selection effects rather than an underlying physical relationship.

6. Birch–Swinnerton-Dyer Independence

The Birch–Swinnerton-Dyer conjecture may provide historical or conceptual motivation for investigating elliptic-curve rank.

It is not a foundational assumption of S.T.A.R.

The research program must remain testable if BSD is removed from the theoretical motivation.

Consequently:

[
 BSD motivation
 $\neq$
 BSD dependence .
]

A mapping or experiment that requires an unproven consequence of BSD must identify that dependency explicitly.

Where possible, the program shall use quantities that can be computed or independently estimated without assuming BSD.

7. Arithmetic Topography

S.T.A.R. may construct an arithmetic scalar field:

$$f(I_1(E), \dots, I_n(E)),$$

where the (I_i) are selected arithmetic invariants.

A corresponding gradient field may be defined:

$$-\nabla h_{\rm arith}.$$

This construction is a candidate arithmetic topography.

It must **not** automatically be identified with gravity.

The correspondence:

$$[\mathbf{F}_{\rm arith} \stackrel{?}{\sim} \mathbf{F}_{\rm gravity}]$$

is an empirical hypothesis requiring independent testing.

Candidate comparisons include:

- matter density;
- gravitational potential;
- weak lensing;
- peculiar velocity;
- radiation fields;
- galaxy morphology;
- star formation rate;

- environmental measurements.
-

8. Cosmic-Web Representation

The cosmic web shall be represented using independently defined astronomical data and appropriate geometric and topological methods.

The principal structural categories are:

- clusters;
- filaments;
- walls/sheets where data support them;
- voids.

Persistent homology is a supporting topological framework, not a replacement for physical or geometric classification.

8.1 (β_0)

[
 β_0
]

represents connected components.

It may therefore provide a topological description related to connected structures and cluster organization.

8.2 (β_1)

[
 β_1
]

represents one-dimensional connectivity/loop structure.

It must **not** automatically be interpreted as “the number of filaments.”

Filament identification shall rely primarily on appropriate geometric, skeleton, graph, or equivalent cosmic-web identification methods.

Persistent homology may then be used as a supporting measure of connectivity, loops, persistence, and structural organization.

This distinction is mandatory.

8.3 (β_2)

[
 β_2
]

represents cavity-like topological structure and may therefore support the analysis of cosmic void topology.

Again, a Betti number is a mathematical descriptor and is not automatically identical to an astronomical physical category.

9. Persistent Homology

For a point cloud or filtered structure

[
 $P_\Phi = \Phi(\mathcal{E})$,
]

S.T.A.R. may construct a filtration:

[
 $K_{\{\epsilon_1\}}$
 $\subseteq$
 $K_{\{\epsilon_2\}}$
 $\subseteq$
 $\cdots$
 $\subseteq$
 $K_{\{\epsilon_n\}}$.
]

Persistent homology then produces persistence diagrams:

$$\begin{bmatrix} D_k(P_{\backslash \Phi}). \end{bmatrix}$$

Corresponding astronomical structures may produce:

$$\begin{bmatrix} D_k(P_{\rm cosmic}). \end{bmatrix}$$

The comparison may use:

$$\begin{bmatrix} W_p(D_k^{\rm arith}, D_k^{\rm cosmic}), \end{bmatrix}$$

or bottleneck distance or another explicitly justified metric.

The distance measure itself does not establish physical correspondence.

10. Historical Numerical Thresholds

Historical numerical criteria must not automatically become scientific acceptance thresholds.

For example, a historical condition such as

$$\begin{bmatrix} W_2 < 10^{-2} \end{bmatrix}$$

shall be classified as:

model-dependent / exploratory

until it has been recalibrated against appropriate null distributions, data resolution, noise, normalization, sample size, mapping alternatives, and parameter-selection effects.

Similarly, historical claims such as:

- (>98%) fidelity;
- (91.9%) correlation;

- (95.6%) correlation;
- fixed Wasserstein thresholds;
- symbolic-regression coefficients;

must be treated as results requiring validation rather than predetermined standards of success.

No threshold shall be considered canonical merely because it appeared in an earlier S.T.A.R. document.

11. Rank–Cosmic-Web Hypothesis Family

The principal rank experiments shall be organized as testable hypotheses.

H1 — Rank–Cluster Hypothesis

Cluster environments exhibit statistically distinguishable rank distributions.

H2 — Rank–Connectivity Hypothesis

Rank correlates with independently measured cosmic-web connectivity.

H3 — Rank–Void Hypothesis

Void environments exhibit a statistically distinguishable arithmetic-invariant distribution.

H4 — Rank–Web Complexity Hypothesis

Arithmetic rank or another arithmetic invariant is associated with independently measured measures of cosmic-web complexity.

H5 — Invariant Competition Hypothesis

Rank is compared against other arithmetic invariants rather than assumed to be uniquely explanatory.

Candidate competitors include:

[
 r_E ,
 quad
 $\text{operatorname{Reg}}(E)$,
 quad

$$\begin{bmatrix} N_E, \\ \quad \quad \quad |\Delta_E|, \\ \quad \quad \quad T_E. \end{bmatrix}$$

A composite vector may be defined as:

$$\begin{bmatrix} X_E = \\ \left(\begin{array}{l} r_E, \\ \log\{\text{Reg}\}_E, \\ \log N_E, \\ \log|\Delta_E|, \\ T_E \end{array} \right). \end{bmatrix}$$

The purpose is to determine whether rank contains information not already explained by other arithmetic quantities.

12. Statistical Requirements

No empirical claim shall be accepted solely because a correlation coefficient, classifier accuracy, visual resemblance, or topological distance appears favorable.

Every substantive empirical claim must address, where applicable:

1. independent data;
2. sample definition;
3. selection effects;
4. null models;
5. randomization;
6. alternative mappings;
7. parameter selection;
8. train/validation/test separation;
9. out-of-sample performance;
10. multiple-hypothesis correction;
11. effect size;
12. uncertainty;

13. robustness to preprocessing;
 14. robustness to resolution and scaling;
 15. replication.
-

13. Null Models

Null models are mandatory.

Appropriate nulls may include:

- randomized arithmetic assignments;
- randomized spatial assignments;
- permutation of rank values;
- shuffled invariant labels;
- synthetic point clouds;
- alternative astronomical catalog realizations;
- simulations without arithmetic information;
- alternative mapping families;
- mappings constructed from mathematically unrelated sequences.

The exact null must be matched to the hypothesis being tested.

A favorable result against an inappropriate null is not sufficient evidence.

14. Alternative Mapping Controls

Whenever feasible, a proposed correspondence shall be tested against multiple mapping constructions.

For example:

```
[  
  \Phi,  
  \quad  
  \Phi_{\rm MCJ},  
  \quad  
  \Phi_{\rm PTD},  
  \quad  
]
```

Φ_{FT} ,
]

may be compared under identical evaluation procedures.

This determines whether an observed result is:

1. invariant across mapping families;
2. specific to one mapping;
3. sensitive to arbitrary parameter choices;
4. consistent with a broader arithmetic principle;
5. or potentially an artifact of the selected construction.

15. Parameter Locking

Parameters used to construct a mapping must be distinguished from parameters discovered during validation.

Parameters selected after observing the target data are potentially subject to overfitting.

Accordingly, S.T.A.R. experiments should distinguish:

[
 $\text{pre-registered/fixed parameters}$
]

from

[
 $\text{fitted/discovered parameters}$.
]

A mapping that is optimized against the same dataset used to evaluate it cannot be treated as an independent prediction.

16. Machine Learning and Symbolic Regression

Machine learning may be used for:

- feature discovery;
- nonlinear relationship discovery;
- classification;
- regression;
- dimensionality reduction;
- hypothesis generation.

Symbolic regression may be used to generate candidate mathematical relationships.

The canonical sequence is:

```
[  
\text{ML discovery}  
\rightarrow  
\text{candidate equation}  
\rightarrow  
\text{independent validation}  
\rightarrow  
\text{analytic derivation}.  
]
```

Machine learning does not constitute mathematical proof.

A symbolic-regression expression is a candidate model until independently validated and, where applicable, mathematically derived.

17. Claim and Evidence Registry

The Claim/Evidence Registry is subordinate to this Charter and is the primary mechanism for tracking the scientific status of S.T.A.R. propositions.

Every significant claim should receive a unique Claim ID.

Each registry entry should contain, where applicable:

1. **Claim ID**
2. **Statement**
3. **Epistemic Category**
4. **Source Document**
5. **Mathematical Basis**

6. **Underlying Assumptions**
7. **Dataset**
8. **Population**
9. **Prediction**
10. **Null Model**
11. **Alternative Models**
12. **Parameters**
13. **Parameter-Locking Status**
14. **Statistical Method**
15. **Effect Size**
16. **Uncertainty**
17. **Result**
18. **Out-of-Sample Result**
19. **Replication Status**
20. **Reproducibility Status**
21. **Dependencies**
22. **Known Limitations**
23. **Contradictory Evidence**
24. **Current Status**
25. **Last Tested**
26. **Code/Experiment Reference**

The Registry shall distinguish between:

claim existence

and

claim support.

The presence of a claim in the Registry does not constitute evidence for the claim.

18. Automatic Reclassification Rules

Historical claims shall be reclassified when their original evidentiary basis does not satisfy the current Charter.

In particular:

18.1 Historical correlations

A historical correlation is classified as **E — preliminary empirical result** only if its analysis can be reproduced and satisfies the applicable statistical requirements.

Otherwise it becomes:

[
\\text{M/Historical Result}
]

or

[
\\text{Exploratory}.
]

18.2 Historical thresholds

Thresholds such as:

[
 $W_2 < 10^{-2}$
]

are automatically classified as:

model-dependent / exploratory

until independently recalibrated.

18.3 Historical symbolic laws

Expressions discovered through symbolic regression are classified as:

candidate models

rather than established laws.

18.4 Historical axioms

Statements previously presented as axioms that assert physical properties of the universe shall be reclassified as:

- hypotheses;
- postulates of a future theory;
- conjectures;
- or speculative interpretations,

depending upon their actual role.

19. Statistical Significance Versus Predictive Utility

S.T.A.R. explicitly distinguishes statistical significance from practical predictive power.

A result may satisfy:

[
 $p < \alpha$
]

and therefore provide evidence against a specified null hypothesis while still having negligible practical predictive utility.

Accordingly:

[
 $\text{statistical significance}$
 $\neq$
 $\text{predictive usefulness}$.
]

A useful predictive model must demonstrate meaningful out-of-sample performance relative to appropriate baselines.

The Registry should therefore record both:

1. statistical evidence of correspondence; and
2. practical predictive utility.

A statistically significant but practically negligible relationship shall not be represented as a successful predictive theory.

20. Cosmic Evolution

Redshift shall initially be treated as an observational ordering variable:

[
z.
]

The program may investigate whether an arithmetic evolution parameter

[
 τ
]

can be related to observations through:

[
 $\tau=f(z)$.
]

The research shall distinguish between:

Topological evolution

Changes in cosmic structural topology with redshift.

Metric expansion

Changes in physical distances and scale factors.

Arithmetic evolution

Changes in an arithmetic representation or arithmetic state parameter.

These must not be assumed to be equivalent.

21. Arithmetic Evolution

A candidate arithmetic state may be represented schematically as:

{ $r_E(\tau)$ }.
]

Its topology may be represented by:

[
 $D_{\rm arithmetic}(\tau).$
]

The observed cosmic topology may be represented by:

[
 $D_{\rm cosmic}(z).$
]

The question is whether there exists a statistically supported correspondence:

[
 $D_{\rm arithmetic}(\tau)$
 $\sim$
 $D_{\rm cosmic}(z)$
]

under an independently justified relation

[
 $\tau=f(z).$
]

This is a hypothesis, not an assumption.

22. Astrophysical Laboratory

Star formation rate provides an independent astrophysical laboratory for testing whether arithmetic or topological quantities contain measurable information about galaxy evolution.

Candidate variables include:

- star formation rate;
- stellar mass;
- gas mass;
- metallicity;
- morphology;
- environment;
- redshift;
- photometric properties;
- structural parameters;

- cosmic-web environment.

The SFR program must remain scientifically meaningful even if the broader arithmetic-cosmic hypothesis is falsified.

Thus:

```
[
\text{SFR correspondence}
\neq
\text{proof of ACSC}.
]
```

It is an independent empirical test.

23. Physical Correspondence

If arithmetic topography demonstrates reproducible astronomical correspondence, S.T.A.R. may investigate whether it corresponds to physical fields.

Candidate comparisons include:

```
[
h_{\rm arith}
\leftrightharpoons
\rho_{\rm matter},
]
```

```
[
\nabla h_{\rm arith}
\leftrightharpoons
\nabla\Phi_{\rm grav},
]
```

and comparisons with:

- weak lensing;
- peculiar velocity;
- radiation;
- galaxy dynamics;
- SFR;
- environmental measures.

No physical identity may be asserted solely from geometric resemblance.

24. Candidate Symbolic Field

A future arithmetic field theory may attempt to construct a field or tensor:

[
 $A_{\mu\nu}$
]

or related object.

The first requirement is mathematical consistency.

Only afterward may one test a correspondence such as:

[
 $A_{\mu\nu}$
 $\stackrel{?}{\sim}$
 $G_{\mu\nu}$.
]

The statement that arithmetic “replaces” the Einstein tensor is not part of the current established S.T.A.R. framework.

It remains a speculative future possibility.

25. Thermodynamic and Mass–Energy Program

S.T.A.R. may investigate whether entropy, cohomology, geodesics, scalar fields, arithmetic invariants, and phase transitions can participate in a generalized thermodynamic or mass–energy relation.

Any proposed relation must satisfy at minimum:

1. dimensional consistency;
2. a mathematically defined action or governing principle where claimed;

3. known physical limits;
4. conservation laws;
5. stability requirements;
6. well-posedness;
7. compatibility with established experimental constraints;
8. independently testable novel predictions.

If a generalized expression contains the standard rest-energy relation, the desired limiting behavior is schematically:

$$[E_{\rm generalized} \rightarrow mc^2]$$

under the appropriate limiting assumptions.

This does not establish that the generalized expression is physically correct.

26. ECC

The Entropy Cohomology Conjecture is treated as a candidate theoretical layer.

The corrected mathematical architecture uses a closed, potentially non-exact target-space two-form:

$$\frac{1}{2}\omega_{AB}(Y)dY^A\wedge dY^B,$$

with

$$[d_{\mathcal{T}}\omega=0, \quad \omega\neq 0.]$$

The corresponding pullback is:

$$\begin{bmatrix} F_Q = \Phi^* \omega. \end{bmatrix}$$

This produces:

$$\begin{bmatrix} dF_Q = 0. \end{bmatrix}$$

Claims concerning entropy/cohomology dynamics remain hypotheses or model constructions unless formally proved and independently validated.

Historical references to a “computational theorem” or “observed symbolic law” shall not be treated as established theorems unless the required mathematical proof and validation exist.

27. RTCH

RTCH is treated as a candidate variational completion rather than a foundational premise.

The candidate framework may contain:

$$\begin{bmatrix} S_{\rm EH} \\ + \\ S_{\phi} \\ + \\ S_Y \\ + \\ S_B \\ + \\ S_{\rm top} \\ + \\ S_m \\ + \\ S_{\rm GHY}. \end{bmatrix}$$

The resulting equations may be investigated for:

- mathematical consistency;
- conservation;
- stability;

- well-posedness;
- known physical limits;
- observational constraints;
- novel predictions.

The existence of a mathematically consistent action does not establish the physical truth of the theory.

28. Falsification

A valid S.T.A.R. hypothesis must admit the possibility of failure.

Examples include:

Mapping failure

Alternative mappings consistently outperform the proposed mapping or the proposed mapping fails against null models.

Rank failure

Rank contains no reproducible information about cosmic-web structure after controlling for relevant confounders.

Topological failure

Arithmetic persistence diagrams do not correspond to astronomical topology beyond null expectations.

Predictive failure

A statistically significant relationship has negligible out-of-sample predictive utility.

Replication failure

The relationship disappears in independent datasets.

Physical failure

A proposed arithmetic field conflicts with established gravitational, cosmological, laboratory, or astrophysical constraints.

Negative results shall be retained in the research archive.

A failed hypothesis is a valid scientific result.

29. Success-Level Ladder

S.T.A.R. uses the following hierarchy.

Level 0 — Mathematical Construction

The mathematical object is precisely defined.

Level 1 — Computational Reproducibility

Independent execution reproduces the result.

Level 2 — Statistical Correspondence

The result demonstrates statistically supported correspondence against appropriate null models.

Level 3 — Cross-Dataset Replication

The correspondence survives independent datasets or independently constructed samples.

Level 4 — Predictive Correspondence

The construction predicts previously unseen data with meaningful out-of-sample utility.

Level 5 — Physical Theory

A mathematically consistent theory produces empirically successful physical predictions while satisfying known constraints.

Importantly:

[
 \text{Level 2}
 \not\rightarrow

\text{Level 4}.
]

Statistical significance alone does not establish practical predictive power.

Likewise:

[
\text{Level 4}
\not\rightarrow
\text{Level 5}.
]

Predictive performance does not by itself establish a fundamental physical theory.

30. Required Separation of Discovery and Confirmation

S.T.A.R. shall distinguish:

[
\text{discovery}
]

from

[
\text{confirmation}.
]

A dataset used to discover a mapping, feature, threshold, equation, or parameter should not simultaneously serve as the sole evidence for confirming that discovery.

The preferred structure is:

[
\text{Discovery}
\rightarrow
\text{Locked Model}
\rightarrow
\text{Blind/Independent Test}]

\rightarrow
\text{Replication}.
]

This applies to:

- machine learning;
- symbolic regression;
- mapping parameters;
- topological thresholds;
- classification boundaries;
- cosmological relationships;
- SFR models.

31. Reproducibility Standard

Every major result should ultimately be associated with:

- source data;
- data version;
- preprocessing;
- code;
- computational environment;
- parameter configuration;
- random seeds where applicable;
- mapping definition;
- statistical procedure;
- null model;
- result artifact;
- experiment identifier.

The goal is that another investigator can reconstruct the result without relying on undocumented choices.

32. Historical Material

Existing S.T.A.R. documents are not discarded merely because their epistemic status changes.

Instead, historical claims are preserved and reclassified.

The principal existing modules are:

ACSC

Principal structural hypothesis concerning arithmetic–cosmic correspondence.

GLMPCT

Mapping and scale-translation framework.

ECC

Candidate entropy/cohomology dynamical layer.

RTCH

Candidate variational completion.

These modules are subordinate to the present Charter.

Where an older document conflicts with this Charter's epistemic standards, the Charter controls the interpretation of the claim.

33. Research Roadmap

The canonical development sequence is:

Phase 0

Research Charter and Claim/Evidence Registry.

Phase 1

Document reconciliation and claim classification.

Phase 2

Arithmetic dataset construction.

Phase 3

Mapping benchmark and alternative mappings.

Phase 4

Arithmetic topography.

Phase 5

Cosmic-web experiment.

Phase 6

Rank experiment.

Phase 7

Cosmic evolution and redshift analysis.

Phase 8

Astrophysical laboratory, including SFR.

Phase 9

Physical correspondence.

Phase 10

Candidate dynamical theory.

Phase 11

Cosmological applications, including the Hubble–Planck tension.

The Hubble–Planck tension is therefore a downstream test rather than a design constraint imposed on the earlier research.

34. Governing Scientific Principle

The governing principle of S.T.A.R. is:

Do not ask whether the theory can be made to fit the universe. Ask whether a fixed mathematical construction predicts structure that was not used to construct it.

This principle controls the relationship between theory development and empirical testing.

A construction that is flexible enough to reproduce an already-used dataset does not thereby establish correspondence.

The scientific objective is instead:

```
[
\boxed{
\text{fixed construction}
\rightarrow
\text{independent prediction}
\rightarrow
\text{independent observation}
}
]
```

with explicit null models and reproducible analysis.

35. Canonical Research Pipeline

The canonical S.T.A.R. pipeline is:

```
[
E
\rightarrow
I(E)
\rightarrow
\Phi(E)
\rightarrow
\mathcal{H}_{\rm arith}
\rightarrow
D_{\rm arith}
\rightarrow
D_{\rm cosmic}
\rightarrow
```

```

\text{Statistical Test}
\rightarrow
\text{Independent Validation}.
]

```

If correspondence survives:

```

[
\text{Correspondence}
\rightarrow
\text{Dynamics}
\rightarrow
\text{Physical Correspondence}
\rightarrow
\text{Candidate Theory}.
]

```

The complete conceptual architecture is therefore:

```

[
\boxed{
\text{Arithmetic}
\rightarrow
\text{Geometry}
\rightarrow
\text{Topography}
\rightarrow
\text{Topology}
\rightarrow
\text{Observation}
\rightarrow
\text{Dynamics}
\rightarrow
\text{Physics}
}
]

```

with every arrow treated as a scientific question rather than an assumed identity.

36. Controlling Standard for the Claim/Evidence Registry

The Claim/Evidence Registry shall use this Charter as its controlling standard.

For every claim, the Registry must answer five fundamental questions:

1. What exactly is being claimed?

The statement must be sufficiently precise to be tested.

2. What is the epistemic status?

The claim must be classified as:

[
D, \ E, \ M, \ P, \ H, \ C, \ T, \ \text{or} \ } S.
]

3. What evidence supports it?

Evidence must be distinguished from assumptions, model outputs, historical observations, and interpretations.

4. What could falsify it?

A claim without a meaningful failure condition shall not be represented as a fully testable empirical hypothesis.

5. Has it survived independent validation?

A result derived from the same data or model used to construct the claim shall not automatically be treated as independent confirmation.

The Registry therefore functions as the scientific memory of S.T.A.R.

It shall preserve:

- positive results;
- negative results;
- unresolved claims;
- competing explanations;
- historical claims;
- superseded interpretations;
- model-dependent thresholds;
- failed experiments;
- successful replications;
- unresolved mathematical questions.

37. Final Standard

S.T.A.R. shall advance claims through the following progression whenever possible:

```
[
\boxed{
\text{Idea}
\rightarrow
\text{Defined Object}
\rightarrow
\text{Hypothesis}
\rightarrow
\text{Prediction}
\rightarrow
\text{Null Test}
\rightarrow
\text{Independent Validation}
\rightarrow
\text{Replication}
\rightarrow
\text{Theory}
}
]
```

No step may be silently skipped.

The purpose of this Charter is not to restrict the scope of S.T.A.R.'s ideas.

Its purpose is to ensure that increasingly ambitious mathematical and physical interpretations remain distinguishable from the evidence supporting them.

The project may ultimately discover:

- a genuine arithmetic–cosmic correspondence;
- a useful mathematical representation with no fundamental physical interpretation;
- an astrophysically useful predictive structure;
- a new mathematical relationship;
- or that particular proposed correspondences do not survive testing.

All of these outcomes are scientifically meaningful.

The controlling objective is therefore not confirmation of ACSC, GLMPCT, ECC, RTCH, rank-based cosmology, or any other predetermined theory.

The controlling objective is the discovery of whatever reproducible mathematical and physical relationships the evidence supports.

[
\\boxed{
\\textbf{S.T.A.R. investigates; the evidence decides.}
}]